Tertiary Infotech Academy Pte. Ltd.

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LESSON PLAN

For

3D Modelling with Blender for Beginners

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Conducted by

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Version 11.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
10.0	Date not recorded in legacy source	Legacy trainer-slide release; superseded by this source-aligned rebuild.	Tertiary Infotech Academy
11.0	20 August 2026	Rebuilt for Blender 5.2 LTS with four verified outcomes, eight connected labs, live assessment parity and updated visuals.	Tertiary Infotech Academy

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Course Overview

A two-day, 16-hour WSQ programme in which beginners interpret a creative brief, build and animate a Blender prototype, then optimise, render and export a verified 3D asset.

Learning Outcomes

- LO1: Analyse a creative brief and establish a suitable polygon budget.
- LO2: Develop 3D models and prototypes in Blender.
- LO3: Animate 3D models using suitable rigging, keyframe and presentation methods.
- LO4: Optimise, light, render and export 3D models for the intended use.

Daily Schedule

Day	Time	Duration	Topic / Lab	Method	Slides
1	09:00–10:00	1h	Course administration and Topic 1 introduction	Briefing, demonstration and discussion	Slides 2–20
1	10:00–12:00	2h	Topic 1: Get Started with Blender and the Creative Brief	Concept workshop and Lab 1	Slides 21–40
1	12:00–13:00	1h	Lunch break	Break	—
1	13:00–15:00	2h	Topic 1 application and Topic 2 introduction	Lab 2 and guided modelling	Slides 41–47
1	15:00–18:00	3h	Topic 2: Develop 3D Models and Prototypes	Demonstration, Labs 3–4 and review	Slides 48–73
2	09:00–10:30	1h 30m	Topic 3: Animate and Present the 3D Prototype	Concept demonstration and coached practice	Slides 74–91
2	10:30–12:30	2h	Topic 3 application	Labs 5–6 and peer review	Slides 92–99
2	12:30–13:30	1h	Lunch break	Break	—
2	13:30–15:00	1h 30m	Topic 4: Optimise, Render and Export the Model	Concept demonstration and Lab 7	Slides 101–122
2	15:00–15:30	0h 30m	Capstone release and assessment briefing	Lab 8 completion and briefing	Slides 123–132
2	15:30–16:30	1h	Written Assessment (SAQ)	Individual, open book	Slides 133–135
2	16:30–18:00	1h 30m	Practical Performance (PP)	Individual, open book	Slides 133–135

Topic-by-Topic Breakdown

Topic 1: Get Started with Blender and the Creative Brief · Slides 19–45

Analyse requirements · establish budgets · control the scene

Alignment: LO1 · K1–K2 · A1–A2

Lab 1: Navigate Blender and organise a production scene · Slides 37–40

Navigate the interface, distinguish Object and Edit modes, organise objects and preserve a clean baseline.

Lab 2: Interpret a fish brief, budget polygons and build a blockout · Slides 41–44

Translate the brief into criteria, allocate a polygon budget and create a modifier-led blockout.

Topic 2: Develop 3D Models and Prototypes · Slides 46–73

Block out form · sculpt detail · refine topology

Alignment: LO2 · K3–K4, K7 · A3–A4

Lab 3: Model and sculpt the production goldfish · Slides 65–68

Develop primary and secondary forms with suitable Edit Mode and sculpting tools.

Lab 4: Repair topology, normals and deformation zones · Slides 69–72

Diagnose and repair the supplied mesh, then prove the revised prototype meets its criteria.

Topic 3: Animate and Present the 3D Prototype · Slides 74–100

Rig movement · keyframe action · surface and present

Alignment: LO3 · K5–K8 · A5–A6

Lab 5: UV unwrap and surface the goldfish · Slides 92–95

Create controlled UVs, textures and a Principled BSDF material that remains readable under neutral light.

Lab 6: Rig and animate a looping swim cycle · Slides 96–99

Build a simple armature, validate deformation and animate a clean loop using controlled curves.

Topic 4: Optimise, Render and Export the Model · Slides 101–127

Frame the asset · validate quality · deliver reliably

Alignment: LO4 · K8–K9 · A7–A8

Lab 7: Create bubbles, light the scene and present for review · Slides 119–122

Add a lightweight bubble/force effect, build purposeful lighting and present a review decision.

Lab 8: Optimise, render and export the fish-bowl capstone · Slides 123–126

Optimise the full scene, render final evidence, export the intended asset and validate the package by round trip.

Resources Required

- Blender 5.2 LTS installed on Windows or macOS laptop
- Course lab folders and supplied Blender sample assets
- Presentation software for lab evidence
- Internet access for official Blender references and LMS submission

Assessment

Written Assessment (SAQ): 9 open-ended questions covering K1–K9; 60 minutes; individual and open book.

Practical Performance (PP): 4 integrated Blender tasks covering A1–A8; 90 minutes; individual and open book.

Total live-paper assessment duration: 150 minutes. Candidate papers are public courseware; answer keys remain trainer-only.

Assessment is individual and open book. Learners complete assessment digital attendance, submit on the LMS and sign the Assessment Summary Record.